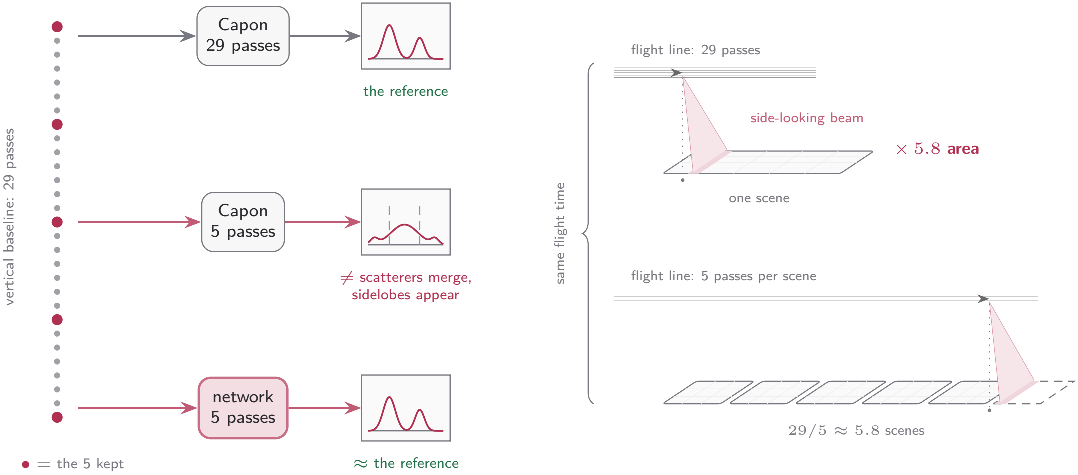
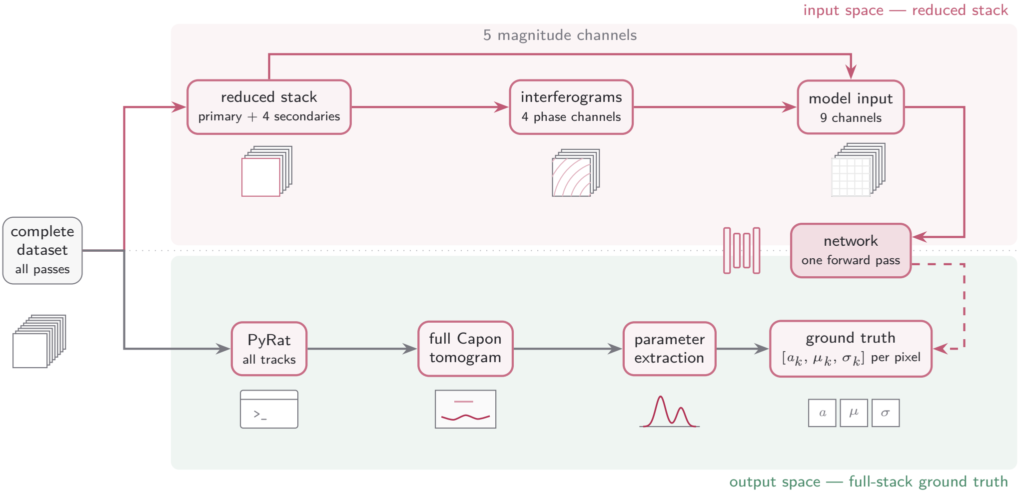
Motivation: more area from the same flight time



My contributions

- **Multi-layered output version**: the same head scaled past two scatterers per pixel $(K > 2)$.
- **Multi-GPU parameter fitting on a JAX framework**: the ground-truth Gaussian fits, batched over the full scene.
- **Improvements in the model architecture**: set-prediction head, dual trunks, the backbone zoo.
- **Improvements in training stability**: matching, the leaky clamp, imbalance levers, sanity gates.
- **Improvements in pre-training**: the per-channel normalization ladder, robust statistics for the heavy-tailed channels.
- **Multiple experiments**: benchmark board, patch sweep, context ladder, input ablations, pair losses.
- **A framework to fully track training development**: launch queue, GPU scheduling, live consoles, progress and alerts.
- **A full toolset for diagnosing and visualizing results**: leaderboards, run health, cube slices, per-pixel microscope.

Methodology



A network learns to reconstruct, from the **reduced** stack alone,
the per-pixel profile parameters that the full-stack tomogram and classical fit produce.

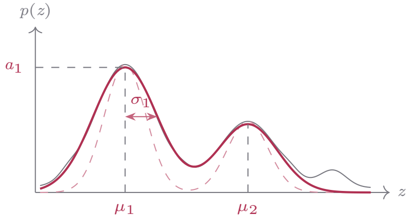
Ground truth: fitted Gaussian parameters

Supervised target: the per-pixel array $[a_k, \mu_k, \sigma_k]_{k=1}^{K}$ ($K{=}2$ today, $K{=}5$ in development), fit to the per-pixel max-normalised Capon spectrum $\tilde{p} = p/p_{\max}$ on the elevation grid ξ_n:

$$\mathcal{L} \;=\; \frac{1}{N}\sum_{n=1}^{N}\big(\hat{p}(\xi_n) - \tilde{p}(\xi_n)\big)^2, \qquad \hat{p}(\xi) = \sum_k a_k\, e^{-\frac{(\xi-\mu_k)^2}{2\sigma_k^2}}$$

One shared loss; each parameter group takes its own projected Adam step:

$$\begin{aligned} a_k &\leftarrow \big[\, a_k - \eta\,\mathrm{Adam}(\partial_{a_k}\mathcal{L})\,\big]_+ \\ \mu_k &\leftarrow \mathrm{clip}_{[\mu^-,\ \mu^+]}\big(\mu_k - \eta\,\mathrm{Adam}(\partial_{\mu_k}\mathcal{L})\big) \\ \sigma_k &\leftarrow \mathrm{clip}_{[\sigma^-,\ \sigma^+]}\big(\sigma_k - \eta\,\mathrm{Adam}(\partial_{\sigma_k}\mathcal{L})\big) \end{aligned}$$



Same pixel: Capon spectrum, peak initialisation (dashed), converged fit; the fitted $[a_k, \mu_k, \sigma_k]$ are the label.

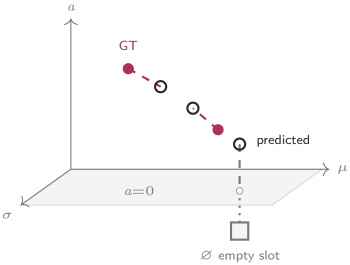
- Fitted in normalised space; the winning amplitudes are denormalised back, $a_k \to p_{\max}\, a_k$ (μ_k, σ_k already live on the physical axis).
- All pixels batched in one optimisation — not a per-pixel loop (next slide); most pixels activate one or two Gaussians (Act III).

Main results

- **Hungarian matching**: assigning predicted scatterers to ground-truth scatterers (next frame).
- **Set-prediction head**: gated slots that decide existence and parameters per scatterer.
- **Receptive field**: how much spatial context the network actually uses.
- **Benchmarking**: 7 backbones $\times$ 5 objectives, five seeds each: the board.
- **Dual backbone**: two trunks, one head: existence and parameters each get their own encoder.
- **Pair loss function**: param-L1 paired with a curve term.

Hungarian matching

- Each Gaussian is a **point in** (a, μ, σ) **space**: prediction and GT are two point sets; the matcher pairs them by distance and scores the *set*, not the slot layout.



Filled = active GT, open = predicted, $\varnothing$ = empty GT slot (below $a=0$). Its match (grey) is zero-weighted, so $\hat\theta_3$ is free.

$$\mathcal{L} = \min_{\pi \in S_K} \sum_{k=1}^{K} w_k \left\| \hat\theta_{\pi(k)} - \theta_k \right\|^2 \qquad \theta_k = (a_k, \mu_k, \sigma_k)$$

$w_k = \mathbf{1}[a_k > \tau]$ — GT *active mask*: 1 if present, 0 if empty.

GT θ_j (columns), pred $\hat\theta_i$ (rows)

	θ_1	θ_2	$\varnothing$
$\hat\theta_1$	0.9	0.2	0.0
$\hat\theta_2$	0.4	0.8	0.0
$\hat\theta_3$	0.6	0.5	0.0

cost $C_{ij} = w_j \|\hat\theta_i - \theta_j\|^2$

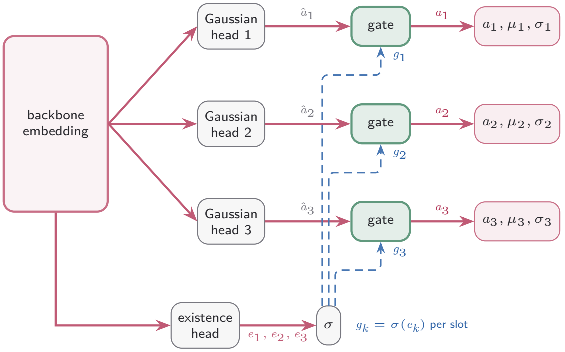
$\xrightarrow[\text{min cost}]{\pi^\star}$

0.9	**0.2**	0.0
0.4	0.8	0.0
0.6	0.5	**0.0**

$\sum C_{\pi^\star} = 0.6$

Column $\varnothing$ is a placeholder GT ($w_j{=}0$) — all-zero, so the leftover prediction matches it for free: $\hat\theta_1 \to \theta_2,\ \hat\theta_2 \to \theta_1,\ \hat\theta_3 \to \varnothing$.

Inside the set-prediction head — gate, blend, match

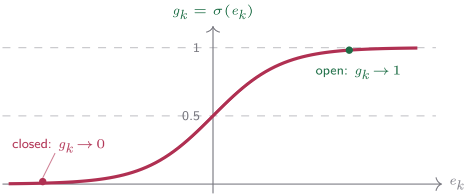


each gate: $a_k = g_k\,\hat{a}_k + (1 - g_k)\,a_{\mathrm{off},k}$

μ_k, σ_k pass through · a_{off} learned per slot

The gate is a sigmoid on the existence logit e_k:

- Open slots $(g_k \to 1)$: $a_k \approx \hat{a}_k$ — the raw regression survives.
- Closed slot $(g_k \to 0)$: $a_k \to a_{\mathrm{off},k} \approx 0$ *whatever* $\hat{a}_k$ is, collapsing onto the empty-set target at near-zero loss.
- e_k is internal — the head still emits $3K$ channels (a_k, μ_k, σ_k); a_{off} is a learned per-slot scalar (init 0).



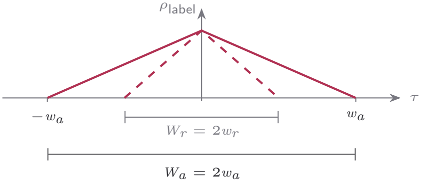
Reading the reach — the network estimates statistics

two labels at offset Δ split into shared and private looks:

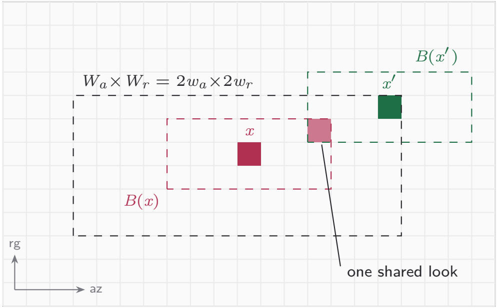
$$w_a w_r \hat{\mathbf{R}}(x) = \sum_{\mathcal{W}\cap\mathcal{W}'} \mathbf{y}_p \mathbf{y}_p^{\mathsf{H}} + \underbrace{\sum_{\mathcal{W}\setminus\mathcal{W}'} \mathbf{y}_p \mathbf{y}_p^{\mathsf{H}}}_{\text{private to } x}$$

$$w_a w_r \hat{\mathbf{R}}(x') = \underbrace{\sum_{\mathcal{W}\cap\mathcal{W}'} \mathbf{y}_p \mathbf{y}_p^{\mathsf{H}}}_{\text{shared } \mathbf{S}(\Delta)} + \underbrace{\sum_{\mathcal{W}'\setminus\mathcal{W}} \mathbf{y}_p \mathbf{y}_p^{\mathsf{H}}}_{\text{private to } x'}$$

the shared fraction at every offset τ is the correlation law:



the label correlates over $\pm w_a$ in azimuth and $\pm w_r$ in range — the centred patch spanning both triangles is $W_a \times W_r$.



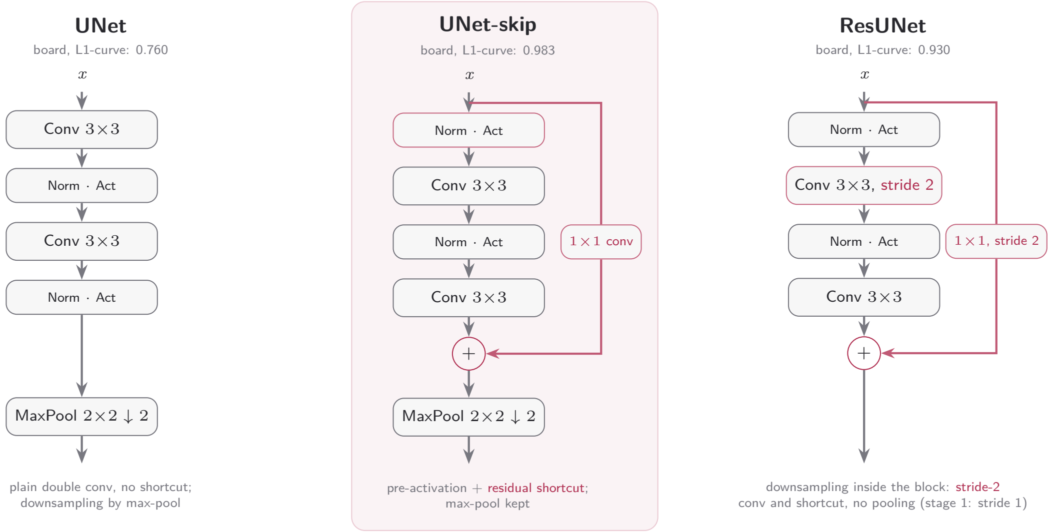
x' sits at the far corner of the patch: $B(x')$ keeps a single shared look with $B(x)$ — one step further and the windows are disjoint ($|\Delta_a| = w_a$ *or* $|\Delta_r| = w_r$).

The board at five seeds — the L1 family owns the top nine

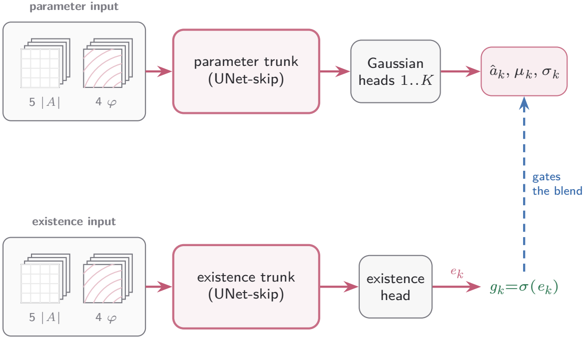
#	model	loss	curve R^2 ↑	cosine ↑	SSIM elev ↑	SSIM range ↑	SSIM azim ↑	peak err ↓	recall ↑	2-sc recall ↑	score
1	unet-skip	L1-curve	**0.756** ±.012	0.924 ±.007	0.9676 ±.0004	0.9500 ±.0016	0.9611 ±.0008	1.29 ±.05	0.886 ±.006	0.728 ±.013	0.983
2	unet-skip	Charbonnier	0.751 ±.012	0.928 ±.004	0.9675 ±.0007	0.9499 ±.0024	0.9608 ±.0012	**1.25** ±.03	0.886 ±.004	0.727 ±.008	0.977
3	unet-skip	param-L1	0.741 ±.016	**0.931** ±.001	0.9669 ±.0004	0.9513 ±.0006	0.9608 ±.0005	1.38 ±.03	0.893 ±.004	0.745 ±.008	0.963
4	resunet	param-L1	0.734 ±.013	0.927 ±.007	0.9664 ±.0003	0.9502 ±.0006	0.9601 ±.0004	1.32 ±.07	0.894 ±.003	**0.746** ±.006	0.946
5	deeplab	param-L1	0.731 ±.006	0.923 ±.004	0.9663 ±.0003	0.9495 ±.0004	0.9597 ±.0003	1.35 ±.05	0.888 ±.002	0.736 ±.005	0.937
6	deeplab	L1-curve	0.731 ±.012	0.921 ±.013	0.9667 ±.0005	0.9491 ±.0016	0.9599 ±.0008	1.33 ±.07	0.873 ±.022	0.721 ±.007	0.935
7	resunet	L1-curve	0.737 ±.019	0.916 ±.007	0.9668 ±.0004	0.9488 ±.0007	0.9601 ±.0006	1.34 ±.06	0.881 ±.005	0.715 ±.007	0.930
8	resunet	Charbonnier	0.731 ±.016	0.921 ±.014	0.9668 ±.0006	0.9492 ±.0015	0.9601 ±.0006	1.31 ±.08	0.881 ±.006	0.716 ±.008	0.929
9	deeplab	Charbonnier	0.735 ±.008	0.913 ±.015	0.9668 ±.0005	0.9492 ±.0020	0.9602 ±.0008	1.62 ±.45	0.868 ±.037	0.722 ±.013	0.912
10	nafnet	Charbonnier	0.702 ±.009	0.906 ±.007	0.9645 ±.0004	0.9444 ±.0010	0.9569 ±.0004	1.48 ±.05	0.875 ±.008	0.711 ±.009	0.829
11	segformer	Charbonnier	0.688 ±.022	0.910 ±.016	0.9651 ±.0008	0.9447 ±.0028	0.9572 ±.0013	1.51 ±.20	0.866 ±.019	0.704 ±.019	0.829
12	nafnet	L1-curve	0.698 ±.007	0.908 ±.010	0.9646 ±.0005	0.9444 ±.0013	0.9568 ±.0007	1.43 ±.08	0.879 ±.005	0.714 ±.008	0.828
						⋮					
16	deeplab	MSE-curve	0.732 ±.010	0.872 ±.019	0.9615 ±.0005	0.9368 ±.0011	0.9527 ±.0004	1.62 ±.08	0.869 ±.004	0.710 ±.003	0.785
20	unet	L1-curve	0.660 ±.015	0.897 ±.026	0.9649 ±.0009	0.9441 ±.0043	0.9563 ±.0018	1.51 ±.28	0.870 ±.029	0.711 ±.020	0.760
21	unet-skip	param-MSE	0.647 ±.047	0.879 ±.017	0.9626 ±.0010	0.9398 ±.0032	0.9536 ±.0016	1.59 ±.13	0.881 ±.022	0.739 ±.014	0.692

runs: model · set-pred · Hungarian · $K{=}2$ · hv flips · presence A · loss · patch 64×32, five seeds; held-out band. Cells: mean ± seed std; **bold** = best mean in column (rows shown), marked only where best and worst differ by more than 5%. score = equal-weight mean of the five metric groups (curve error, variance, SSIM, cosine, peak), min–max scaled over the 35 arms; # = rank; ranks 13–15, 17–19, 22–35 on the next frame only. cosine = per-pixel profile cosine mean; peak err in elevation units; SSIM per axis vs the GT cube (denorm); recall = matched, 2-sc recall on GT $k{=}2$ pixels. green = the winner's score; red = the param-MSE reference cell, 21st of 35.

One encoder block: UNet vs UNet-skip vs ResUNet



The dual model — two trunks, one head



each gate: $a_k = g_k\,\hat{a}_k + (1 - g_k)\,a_{\mathrm{off},k}$

the set-prediction head of Act III, unchanged — only the source of each embedding splits

- **Split the encoder, keep the head**: two independent UNet-skip trunks — the Gaussian heads read one embedding, the existence head the other. Gate blend and Hungarian matching stay as they are.
- **Each trunk picks its own stack**: the channel groups $|A|$ (5 amplitudes) and φ (4 interferograms) are selected per trunk — which modality reaches which head becomes an architectural choice.
- **Why route**: the input ablation split the labour — phase fires the slots, the full stack calibrates the parameters. The dual model can hand each head the input its job needs.

Pairing the objective — param-L1 plus a curve term

$$\mathcal{L}(w) \;=\; \left(\mathcal{L}_{\text{param-L1}} \;+\; w\,\mathcal{L}_{\text{curve}}\right) \,/\, (1+w)$$

paired weight w

L1	0.01	0.05	0.25	0.5	1	magnitude
Charbonnier	0.01	0.05	0.25	0.5	1	magnitude
cosine	0.01	0.05	0.25	0.5	1	shape only

baseline — param-L1 alone (w=0)

1 baseline + 15 pair arms per K, five seeds each, at K=2 and K=5 = 160 runs

The ratio block left the K=5 gate under-firing at every split — more existence capacity did not wake it. This experiment changes the objective instead: keep param-L1 and add a synthesized-curve term.

- **Two signals, one loss**: param-L1 reaches a slot only through its matched pair; the curve term re-synthesizes the profile from *all* slots and is permutation-invariant — its gradient touches every slot, matched or not.
- **Magnitude or shape**: L1 and Charbonnier price the synthesized profile's magnitude; cosine normalizes both curves first and prices shape alone.
- **A normalized pair**: the two terms are averaged, so the validation loss changes meaning with w — arms compare on held-out metrics only, never on the loss.
- **The question**: if the conservative gate is starved of a profile-level error signal rather than capacity, a small w should move detection at K=5 and leave K=2 alone.

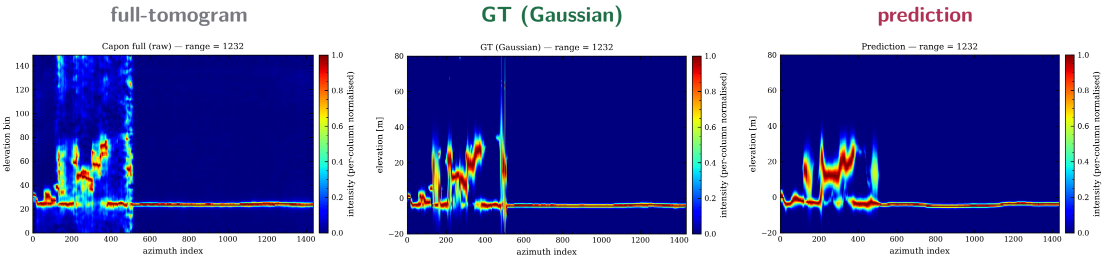
The pair winner, measured: param-L1 + 0.5 cosine

metric	value
training (validation split)	
best val loss ↓	0.1565 ±.0010
best epoch (val)	47
reconstruction (curve)	
curve RMSE ↓	0.214 ±.004
curve MAE ↓	0.0364 ±.0002
curve R^2 ↑	0.754 ±.008
per-pixel R^2 (mean) ↑	0.289 ±.011
PSNR (dB) ↑	52.76 ±.15
profile cosine ↑	0.937 ±.002
structure (vs GT, denorm)	
SSIM elev ↑	0.9675 ±.0002
SSIM range ↑	0.9518 ±.0002
SSIM azim ↑	0.9616 ±.0002
peak location	
peak err mean ↓	1.205 ±.005
peak err p95 ↓	4.70 ±0

metric	value
detection (matched)	
recall ↑	0.902 ±.002
k=1	0.999 ±.000
k=2	0.764 ±.005
precision ↑	0.891 ±.008
k=1	0.930 ±.012
k=2	0.827 ±.004
F1 ↑	0.897 ±.004
count exact ↑	0.905 ±.010
component error (matched pairs)	
μ MAE ↓	1.691 ±.036
k=1	0.345 ±.007
k=2	3.763 ±.085
σ MAE ↓	0.765 ±.005
k=1	0.199 ±.003
k=2	1.635 ±.017
a MAE ↓	0.182 ±.003
k=1	0.059 ±.002
k=2	0.372 ±.004

run: dual UNet-skip trunks · set-pred · Hungarian · K=2 · hv flips · presence A · full stack → both trunks · loss = param-L1 + 0.5 cosine curve; five seeded replicas, held-out band, cells mean ± seed std. SSIM per axis vs the GT cube (denorm), at 4 decimals (seed spreads ≤ 0.0004). k = true scatterer count of the pixel; μ, σ and peak error in elevation units, a in linear amplitude; recall/precision/F1 matched.

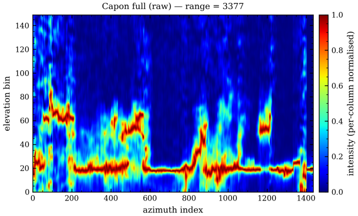
What the network reconstructs: range cut 1232



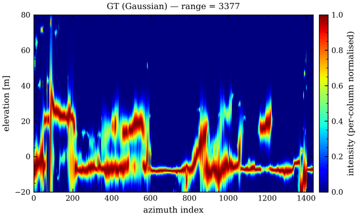
Constant-range cut $\mathrm{rg} = 1232$ through the cube: azimuth on the horizontal axis, elevation on the vertical, every azimuth column normalised to its own peak. **Left**: the raw full-stack Capon tomogram the labels are fitted from, elevation in inversion-grid bins. **Middle**: the Gaussian mixture re-synthesised from the GT parameters, $z \in [-20, 80]$ m. **Right**: the same mixture re-synthesised from the pair winner's predicted parameters (param-L1 $+ 0.5$ cosine).

What the network reconstructs: range cut 3377

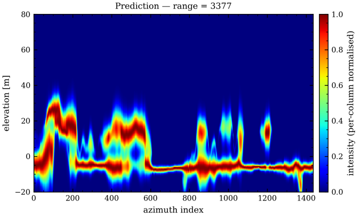
full-tomogram



GT (Gaussian)



prediction



The second cut, $\mathrm{rg} = 3377$, same three panels: raw full-stack tomogram in inversion-grid bins, GT re-synthesis and prediction re-synthesis in $z \in [-20, 80]$ m, per-column peak normalisation.

Next steps

- **Physical loss terms**: the five ranked terms and the first single-term $K{=}2$ sweep results on the next eight frames; the Capon-cycle arm is still queued.
- γ**-net adaptation**: adapting the γ-net architecture to this pipeline.
- **JEPA pretraining**: predicting in representation space, the closing frame.

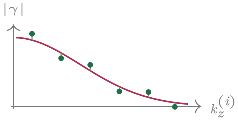
Data-consistency terms (the ranked top two)

Coherence re-synthesis

coherence domain

$$\gamma_P(k_z^{(i)}) = \frac{\sum_n P(\xi_n)\, e^{jk_z^{(i)}\xi_n}}{\sum_n P(\xi_n)}$$

$$\ell_{\text{coh}} = \left\langle \frac{1}{N_s} \sum_i \left|\gamma_P(k_z^{(i)}) - \gamma_T(k_z^{(i)})\right|^2 \right\rangle$$



γ_P from the predicted mixture pulled onto the target coherences, baseline by baseline.

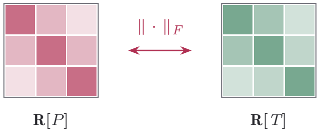
Cloude 2006 (PCT); same Fourier kernel as gamma-Net (Qian et al.).

Covariance matching

covariance domain

$$\mathbf{R}[P] = \mathbf{A}\,\mathrm{diag}(P)\,\mathbf{A}^H \Delta\xi$$

$$\ell_{\text{cov}} = \left\langle \frac{\|\mathbf{R}[P]-\mathbf{R}[T]\|_F^2}{\|\mathbf{R}[T]\|_F^2} \right\rangle$$



Synthesised vs target covariance — structure and relative scale, entry by entry.

COMET (Ottersten, Stoica, Roy 1998); TomoSAR: Cros & Ferro-Famil 2025; CS fidelity: Berenger et al. 2023.

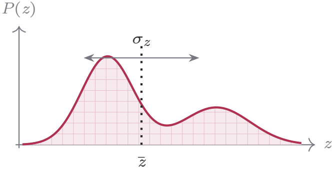
Descriptor terms: moments and total power

Moments

mass · phase-centre $\bar{z}$ · spread σ_z

$$m = \sum_n P(\xi_n)\,\Delta\xi$$

$$\bar{z} = \frac{\sum_n \xi_n P(\xi_n)}{\sum_n P(\xi_n)}, \qquad \sigma_z^2 = \frac{\sum_n (\xi_n - \bar{z})^2 P(\xi_n)}{\sum_n P(\xi_n)}$$



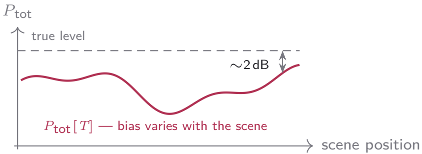
Centroid and spread are bias-robust *ratios* (Cros & Ferro-Famil 2025).

Total power

the one absolute-power quantity

$$P_{\text{tot}}[\,\cdot\,] = \sum_n (\,\cdot\,)(\xi_n)\,\Delta\xi$$

$$\ell_{\text{pow}} = \left\langle \left(P_{\text{tot}}[P] - P_{\text{tot}}[T]\right)^2 \right\rangle$$

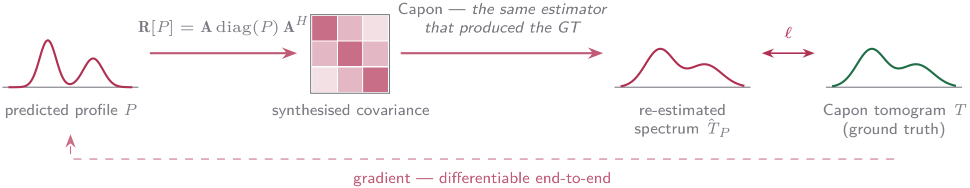


The target $P_{\text{tot}}[T]$ is not the true power: its bias moves with scene content (Lopez-Dekker & Mallorqui 2010) — the loss would imprint that bias.

The estimator-cycle term: Capon cycle

Push the prediction through the *same estimator* that produced the ground truth:

$$\hat{T}_P(\xi_n) = \left(\mathbf{a}^H(\xi_n)(\mathbf{R}[P] + \epsilon\bar{\sigma}\mathbf{I})^{-1}\mathbf{a}(\xi_n)\right)^{-1}$$



- Absorbs the Capon bias and the PSF **by construction**.
- Costs an $N_s \times N_s$ solve per pixel — the most expensive of the five.

Cycle-consistency lineage: Yaman 2020, Lu 2021, Vella & Mota 2020, Angermann 2023.

The physics sweep at $K=2$ — every term rides low, the scale terms collapse high

	reconstruction		detection (matched)						
objective	curve R^2 ↑	cos med ↑	prec. ↑	recall ↑	F1 ↑	exact ↑	under ↓	over ↓	peak p95 ↓
param-L1 + cos	0.747 ±.006	0.965 ±.004	0.895 ±.008	0.901 ±.001	0.898 ±.004	0.908 ±.007	0.041 ±.003	0.051 ±.009	4.7 ±.0
+ *coherence re-synthesis*									
w=0.01	0.752 ±.004	0.966 ±.002	0.895 ±.006	0.902 ±.001	0.898 ±.003	0.910 ±.002	0.040 ±.004	0.050 ±.005	4.7 ±.0
w=0.05	0.752 ±.006	0.966 ±.001	0.893 ±.005	0.902 ±.001	0.898 ±.002	0.911 ±.003	0.039 ±.003	0.051 ±.004	4.7 ±.0
w=0.25	0.755 ±.007	0.966 ±.002	0.892 ±.009	0.903 ±.002	0.897 ±.004	0.909 ±.005	0.038 ±.005	0.053 ±.010	4.7 ±.0
w=0.5	0.748 ±.009	0.965 ±.003	0.886 ±.013	0.903 ±.001	0.895 ±.007	0.911 ±.010	0.032 ±.006	0.057 ±.014	4.7 ±.0
w=1	0.751 ±.008	0.966 ±.002	0.870 ±.045	0.903 ±.001	0.886 ±.024	0.885 ±.063	0.032 ±.005	0.083 ±.068	4.7 ±.0
+ *covariance matching*									
w=0.01	0.752 ±.001	0.965 ±.002	0.896 ±.005	0.901 ±.001	0.898 ±.003	0.906 ±.004	0.043 ±.001	0.051 ±.005	4.7 ±.0
w=0.05	0.748 ±.001	0.966 ±.002	0.899 ±.005	0.902 ±.001	0.900 ±.002	0.909 ±.004	0.043 ±.002	0.048 ±.005	4.7 ±.0
w=0.25	0.738 ±.004	0.966 ±.002	0.896 ±.006	0.899 ±.002	0.898 ±.003	0.902 ±.008	0.045 ±.002	0.053 ±.008	4.7 ±.0
w=0.5	0.727 ±.011	0.965 ±.002	0.900 ±.001	0.898 ±.002	0.899 ±.001	0.907 ±.003	0.046 ±.003	0.047 ±.001	4.7 ±.0
w=1	0.723 ±.006	0.965 ±.001	0.899 ±.005	0.899 ±.002	0.899 ±.002	0.903 ±.003	0.047 ±.004	0.050 ±.006	4.7 ±.0
+ *moments*									
w=0.01	0.753 ±.003	0.966 ±.002	0.895 ±.006	0.902 ±.001	0.898 ±.003	0.910 ±.004	0.040 ±.005	0.050 ±.006	4.7 ±.0
w=0.05	0.755 ±.002	0.966 ±.002	0.896 ±.005	0.902 ±.001	0.899 ±.002	0.910 ±.004	0.041 ±.004	0.049 ±.005	4.7 ±.0
w=0.25	0.751 ±.002	0.965 ±.002	0.897 ±.004	0.901 ±.001	0.899 ±.002	0.908 ±.003	0.044 ±.000	0.049 ±.004	4.7 ±.0
w=0.5	0.750 ±.004	0.966 ±.002	0.898 ±.003	0.900 ±.001	0.899 ±.001	0.909 ±.001	0.044 ±.002	0.047 ±.003	4.7 ±.0
w=1	0.746 ±.005	0.966 ±.001	0.897 ±.003	0.901 ±.001	0.899 ±.002	0.908 ±.003	0.043 ±.001	0.049 ±.003	4.7 ±.0
+ *total power*									
w=0.01	0.752 ±.003	0.965 ±.002	0.896 ±.005	0.902 ±.001	0.899 ±.002	0.910 ±.004	0.041 ±.004	0.049 ±.005	4.7 ±.0
w=0.05	0.751 ±.005	0.966 ±.001	0.897 ±.003	0.900 ±.001	0.899 ±.002	0.908 ±.002	0.044 ±.001	0.048 ±.002	4.7 ±.0
w=0.25	0.747 ±.003	0.965 ±.001	0.898 ±.003	0.900 ±.001	0.899 ±.001	0.908 ±.004	0.044 ±.003	0.049 ±.004	4.7 ±.0
w=0.5	0.737 ±.015	0.965 ±.002	0.900 ±.005	0.899 ±.002	0.899 ±.003	0.905 ±.006	0.048 ±.004	0.047 ±.006	4.7 ±.0
w=1	0.724 ±.020	0.963 ±.005	0.902 ±.003	0.898 ±.005	0.900 ±.001	0.907 ±.003	0.049 ±.007	**0.044** ±.004	4.8 ±.3

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\to$ both trunks) · parity twins. Row = loss (param-L1 + 0.5 cos + w phys) / $(1.5 + w)$, one physics term per arm; param-L1 + cos = the no-physics reference. Five seeded replicas per arm, cells mean ± seed std; held-out test, 5.0 M px. Val losses do not compare across arms (different composites); curve MAE / PSNR / SSIM stay flat outside the collapse arms (next frames); peak p95 is pinned at 4.7. **bold** = best in column, marked only where best and worst differ by more than 5%. Green: coherence $w=0.25$ takes the best curve R^2; at $w=1$ its gate trades misses for false alarms (under 0.041 $\to$ 0.032, over 0.051 $\to$ 0.083; seed lottery: precision ±.045, exact ±.063). Red = absolute-scale collapse past $w=0.25$ (covariance, total power). cos med: median per-pixel cosine to GT; peak p95: 95th pct of the per-pixel peak error.

The $K=2$ physics detection by scatterer count — the set is untouched at every weight

objective	precision ↑ all	precision ↑ $k=1$	precision ↑ $k=2$	recall ↑ all	recall ↑ $k=1$	recall ↑ $k=2$	count acc | pred k ↑ $k=1$	count acc | pred k ↑ $k=2$
param-L1 + cos	0.895 ±.008	0.935 ±.010	0.828 ±.006	0.901 ±.001	0.999 ±.000	0.763 ±.002	0.944 ±.004	0.814 ±.024
+ *coherence re-synthesis*								
$w=0.01$	0.895 ±.006	0.935 ±.006	0.828 ±.006	0.902 ±.001	0.999 ±.000	0.765 ±.001	0.945 ±.005	0.815 ±.012
$w=0.05$	0.893 ±.005	0.935 ±.005	0.826 ±.005	0.902 ±.001	0.999 ±.000	0.765 ±.002	0.947 ±.004	0.814 ±.011
$w=0.25$	0.892 ±.009	0.932 ±.011	0.826 ±.006	0.903 ±.002	0.999 ±.000	0.766 ±.004	0.948 ±.006	0.808 ±.023
$w=0.5$	0.886 ±.013	0.928 ±.016	0.818 ±.010	0.903 ±.001	0.999 ±.000	0.767 ±.003	0.955 ±.008	0.803 ±.033
$w=1$	0.870 ±.045	0.902 ±.067	0.818 ±.009	0.903 ±.001	0.999 ±.000	0.768 ±.001	0.954 ±.002	0.755 ±.122
+ *covariance matching*								
$w=0.01$	0.896 ±.005	0.935 ±.006	0.831 ±.004	0.901 ±.001	0.999 ±.000	0.761 ±.002	0.943 ±.003	0.812 ±.016
$w=0.05$	0.899 ±.005	0.938 ±.006	0.834 ±.003	0.902 ±.001	0.999 ±.000	0.763 ±.003	0.943 ±.002	0.819 ±.015
$w=0.25$	0.896 ±.006	0.932 ±.009	0.835 ±.002	0.899 ±.002	0.999 ±.000	0.758 ±.005	0.944 ±.003	0.803 ±.023
$w=0.5$	0.900 ±.001	0.939 ±.002	0.836 ±.002	0.898 ±.002	0.999 ±.000	0.755 ±.005	0.943 ±.005	0.820 ±.005
$w=1$	0.899 ±.005	0.936 ±.007	0.837 ±.002	0.899 ±.002	0.999 ±.000	0.756 ±.005	0.942 ±.007	0.812 ±.015
+ *moments*								
$w=0.01$	0.895 ±.006	0.936 ±.007	0.827 ±.007	0.902 ±.001	0.999 ±.000	0.764 ±.002	0.945 ±.006	0.816 ±.015
$w=0.05$	0.896 ±.005	0.937 ±.006	0.828 ±.006	0.902 ±.001	0.999 ±.000	0.764 ±.003	0.944 ±.005	0.817 ±.015
$w=0.25$	0.897 ±.004	0.937 ±.004	0.831 ±.004	0.901 ±.001	0.999 ±.000	0.761 ±.003	0.941 ±.001	0.817 ±.011
$w=0.5$	0.898 ±.003	0.939 ±.004	0.832 ±.003	0.900 ±.001	0.999 ±.000	0.761 ±.003	0.942 ±.002	0.821 ±.009
$w=1$	0.897 ±.003	0.937 ±.004	0.832 ±.003	0.901 ±.001	0.999 ±.000	0.761 ±.002	0.943 ±.001	0.817 ±.009
+ *total power*								
$w=0.01$	0.896 ±.005	0.937 ±.006	0.828 ±.006	0.902 ±.001	0.999 ±.000	0.763 ±.001	0.944 ±.005	0.818 ±.013
$w=0.05$	0.897 ±.003	0.938 ±.003	0.831 ±.004	0.900 ±.001	0.999 ±.000	0.761 ±.003	0.941 ±.001	0.819 ±.007
$w=0.25$	0.898 ±.003	0.937 ±.004	0.833 ±.004	0.900 ±.001	0.999 ±.000	0.761 ±.003	0.942 ±.002	0.817 ±.011
$w=0.5$	0.900 ±.005	0.939 ±.007	0.834 ±.003	0.899 ±.002	0.999 ±.000	0.757 ±.006	0.937 ±.006	0.820 ±.018
$w=1$	0.902 ±.003	0.943 ±.005	0.834 ±.002	0.898 ±.005	0.999 ±.000	0.755 ±.011	0.934 ±.008	**0.829** ±.009

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack → both trunks) · parity twins. Row = one physics term at weight w; param-L1 + cos = the no-physics reference. Five seeded replicas per arm, cells mean ± seed std; held-out test, 5.0 M px. All rows matched (Hungarian pairing); k = true scatterer count of the pixel. **bold** = best in column, marked only where best and worst differ by more than 5% — almost nothing clears it at any weight: the physics terms shape the profile, not the scatterer set. The exception is the high-weight coherence gate: $k=2$ recall creeps up (0.763 → 0.768) while the claimed pairs destabilise ($k=1$ precision seed lottery ±.067, pred $k=2$ accuracy ±.122). Precision rewards under-prediction — rank on recall. count acc | pred k: share of pixels claiming exactly k scatterers whose true count is k.

The $K{=}2$ physics component errors — covariance buys position, coherence pays it

objective	μ MAE ↓			σ MAE ↓			a MAE ↓		
	all	$k{=}1$	$k{=}2$	all	$k{=}1$	$k{=}2$	all	$k{=}1$	$k{=}2$
param-L1 + cos	1.71 ±.07	0.36 ±.01	3.78 ±.14	0.774 ±.011	0.193 ±.002	1.668 ±.023	0.182 ±.004	0.058 ±.003	0.373 ±.006
+ *coherence re-synthesis*									
$w{=}0.01$	1.70 ±.06	0.35 ±.01	3.78 ±.12	0.770 ±.009	0.194 ±.003	1.654 ±.024	**0.180** ±.003	**0.057** ±.002	0.369 ±.004
$w{=}0.05$	1.72 ±.05	0.35 ±.01	3.82 ±.11	0.779 ±.014	0.195 ±.002	1.673 ±.035	0.180 ±.003	0.057 ±.003	**0.369** ±.004
$w{=}0.25$	1.72 ±.06	0.35 ±.00	3.82 ±.13	0.778 ±.011	0.192 ±.003	1.675 ±.016	0.182 ±.005	0.058 ±.004	0.373 ±.005
$w{=}0.5$	1.80 ±.10	0.35 ±.01	4.00 ±.22	0.804 ±.017	0.191 ±.002	1.730 ±.033	0.183 ±.004	0.058 ±.003	0.372 ±.004
$w{=}1$	1.81 ±.12	0.35 ±.00	4.02 ±.26	0.809 ±.020	0.194 ±.006	1.737 ±.035	0.185 ±.006	0.061 ±.008	0.373 ±.004
+ *covariance matching*									
$w{=}0.01$	1.66 ±.04	0.35 ±.01	3.70 ±.09	0.755 ±.014	0.192 ±.004	1.626 ±.032	0.182 ±.003	0.058 ±.003	0.374 ±.004
$w{=}0.05$	1.65 ±.02	0.35 ±.01	3.66 ±.05	0.754 ±.007	0.194 ±.002	1.620 ±.017	0.187 ±.005	0.061 ±.004	0.381 ±.007
$w{=}0.25$	1.62 ±.01	0.34 ±.01	3.62 ±.04	0.725 ±.011	0.186 ±.002	**1.566** ±.024	0.196 ±.004	0.065 ±.004	0.400 ±.006
$w{=}0.5$	**1.61** ±.01	0.34 ±.01	**3.61** ±.03	**0.724** ±.014	**0.186** ±.003	1.567 ±.035	0.205 ±.005	0.069 ±.004	0.418 ±.010
$w{=}1$	1.61 ±.02	0.35 ±.01	3.61 ±.04	0.728 ±.018	0.189 ±.008	1.575 ±.039	0.214 ±.012	0.071 ±.005	0.439 ±.024
+ *moments*									
$w{=}0.01$	1.71 ±.06	0.35 ±.01	3.79 ±.14	0.772 ±.017	0.195 ±.003	1.657 ±.041	0.180 ±.003	0.057 ±.002	0.369 ±.004
$w{=}0.05$	1.69 ±.04	0.35 ±.01	3.75 ±.09	0.771 ±.018	0.195 ±.003	1.657 ±.043	0.182 ±.003	0.058 ±.003	0.372 ±.002
$w{=}0.25$	1.68 ±.04	0.35 ±.01	3.72 ±.09	0.763 ±.005	0.195 ±.003	1.643 ±.015	0.186 ±.004	0.060 ±.003	0.380 ±.005
$w{=}0.5$	1.66 ±.03	0.35 ±.01	3.69 ±.07	0.759 ±.007	0.194 ±.002	1.634 ±.016	0.190 ±.004	0.062 ±.004	0.387 ±.005
$w{=}1$	1.66 ±.02	0.35 ±.01	3.69 ±.04	0.760 ±.008	0.196 ±.004	1.633 ±.012	0.192 ±.002	0.063 ±.003	0.391 ±.003
+ *total power*									
$w{=}0.01$	1.70 ±.05	0.35 ±.01	3.77 ±.10	0.771 ±.019	0.196 ±.004	1.656 ±.045	0.180 ±.003	0.057 ±.003	0.370 ±.003
$w{=}0.05$	1.67 ±.03	0.35 ±.00	3.72 ±.08	0.761 ±.005	0.194 ±.003	1.639 ±.015	0.184 ±.004	0.059 ±.004	0.377 ±.004
$w{=}0.25$	1.65 ±.03	0.35 ±.01	3.66 ±.05	0.753 ±.011	0.195 ±.004	1.619 ±.016	0.188 ±.005	0.060 ±.005	0.385 ±.006
$w{=}0.5$	1.63 ±.02	0.35 ±.01	3.62 ±.04	0.762 ±.022	0.197 ±.006	1.646 ±.052	0.192 ±.004	0.062 ±.003	0.396 ±.007
$w{=}1$	1.63 ±.02	0.35 ±.01	3.63 ±.03	0.777 ±.021	0.205 ±.008	1.672 ±.055	0.198 ±.005	0.063 ±.003	0.410 ±.012

runs: dual UNet-skip trunks · set-pred · Hungarian · $K{=}2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack → both trunks) · parity twins. Row = one physics term at weight w; param-L1 + cos = the no-physics reference. Five seeded replicas per arm, cells mean ± seed std; held-out test, $5.0\,$M px. Component errors over matched pairs only (Hungarian pairing); k = true scatterer count of the pixel. **bold** = best in column, marked only where best and worst differ by more than 5%. Covariance at $w{=}0.05$ takes the best μ (1.65 vs 1.71); full-weight coherence pays position and width (μ 1.81, σ 0.809) — the wider gate matches harder pairs; amplitude never moves. μ, σ in elevation units; a in linear amplitude.

The $K{=}2$ physics curve fidelity — flat everywhere except the collapse arms

	curve			SSIM vs GT			peak
objective	MAE ↓	RMSE ↓	PSNR [dB] ↑	elev ↑	range ↑	azim ↑	mean ↓
param-L1 + cos	0.0370 $\pm$.0005	0.217 $\pm$.002	52.63 $\pm$.10	0.9671 $\pm$.0004	0.9510 $\pm$.0008	0.9610 $\pm$.0006	1.23 $\pm$.02
+ *coherence re-synthesis*							
$w{=}0.01$	0.0367 $\pm$.0003	0.215 $\pm$.002	52.71 $\pm$.07	0.9673 $\pm$.0002	0.9513 $\pm$.0006	0.9613 $\pm$.0004	1.22 $\pm$.01
$w{=}0.05$	0.0368 $\pm$.0003	0.215 $\pm$.003	52.72 $\pm$.10	0.9673 $\pm$.0002	0.9512 $\pm$.0005	0.9612 $\pm$.0004	1.22 $\pm$.01
$w{=}0.25$	0.0366 $\pm$.0003	**0.213** $\pm$.003	52.78 $\pm$.13	0.9674 $\pm$.0002	0.9516 $\pm$.0004	0.9615 $\pm$.0003	1.22 $\pm$.01
$w{=}0.5$	0.0370 $\pm$.0006	0.217 $\pm$.004	52.65 $\pm$.16	0.9670 $\pm$.0006	0.9504 $\pm$.0015	0.9607 $\pm$.0010	1.23 $\pm$.02
$w{=}1$	0.0371 $\pm$.0007	0.215 $\pm$.003	52.70 $\pm$.14	0.9670 $\pm$.0006	0.9499 $\pm$.0027	0.9606 $\pm$.0014	1.22 $\pm$.01
+ *covariance matching*							
$w{=}0.01$	0.0366 $\pm$.0001	0.215 $\pm$.001	52.72 $\pm$.02	0.9675 $\pm$.0002	0.9520 $\pm$.0004	0.9615 $\pm$.0003	1.22 $\pm$.01
$w{=}0.05$	0.0365 $\pm$.0002	0.217 $\pm$.000	52.65 $\pm$.02	0.9675 $\pm$.0002	0.9521 $\pm$.0004	0.9616 $\pm$.0002	1.21 $\pm$.01
$w{=}0.25$	0.0368 $\pm$.0002	0.221 $\pm$.002	52.49 $\pm$.06	0.9673 $\pm$.0001	0.9516 $\pm$.0003	0.9612 $\pm$.0002	1.23 $\pm$.01
$w{=}0.5$	0.0375 $\pm$.0007	0.225 $\pm$.005	52.31 $\pm$.18	0.9669 $\pm$.0005	0.9509 $\pm$.0008	0.9606 $\pm$.0007	1.23 $\pm$.01
$w{=}1$	0.0377 $\pm$.0005	0.227 $\pm$.002	52.24 $\pm$.09	0.9670 $\pm$.0003	0.9510 $\pm$.0006	0.9605 $\pm$.0005	1.24 $\pm$.01
+ *moments*							
$w{=}0.01$	0.0366 $\pm$.0002	0.215 $\pm$.001	52.73 $\pm$.05	0.9674 $\pm$.0002	0.9516 $\pm$.0006	0.9614 $\pm$.0004	1.22 $\pm$.01
$w{=}0.05$	0.0365 $\pm$.0002	0.214 $\pm$.001	52.76 $\pm$.04	0.9674 $\pm$.0002	0.9519 $\pm$.0004	0.9615 $\pm$.0003	1.22 $\pm$.01
$w{=}0.25$	0.0366 $\pm$.0002	0.215 $\pm$.001	52.71 $\pm$.04	0.9674 $\pm$.0002	0.9519 $\pm$.0004	0.9615 $\pm$.0004	1.22 $\pm$.01
$w{=}0.5$	0.0365 $\pm$.0003	0.216 $\pm$.002	52.69 $\pm$.07	0.9675 $\pm$.0002	0.9520 $\pm$.0005	0.9616 $\pm$.0004	1.22 $\pm$.01
$w{=}1$	0.0367 $\pm$.0003	0.218 $\pm$.002	52.61 $\pm$.08	0.9675 $\pm$.0003	0.9519 $\pm$.0004	0.9614 $\pm$.0003	1.22 $\pm$.01
+ *total power*							
$w{=}0.01$	0.0367 $\pm$.0002	0.215 $\pm$.001	52.71 $\pm$.06	0.9673 $\pm$.0002	0.9517 $\pm$.0005	0.9614 $\pm$.0004	1.22 $\pm$.00
$w{=}0.05$	0.0366 $\pm$.0001	0.215 $\pm$.002	52.70 $\pm$.08	0.9674 $\pm$.0001	0.9519 $\pm$.0004	0.9615 $\pm$.0003	1.22 $\pm$.01
$w{=}0.25$	0.0367 $\pm$.0003	0.217 $\pm$.001	52.64 $\pm$.05	0.9674 $\pm$.0002	0.9519 $\pm$.0005	0.9614 $\pm$.0004	1.23 $\pm$.01
$w{=}0.5$	0.0373 $\pm$.0011	0.221 $\pm$.006	52.48 $\pm$.24	0.9669 $\pm$.0009	0.9512 $\pm$.0013	0.9608 $\pm$.0011	1.22 $\pm$.01
$w{=}1$	0.0377 $\pm$.0015	0.227 $\pm$.008	52.27 $\pm$.31	0.9667 $\pm$.0011	0.9508 $\pm$.0016	0.9605 $\pm$.0014	1.23 $\pm$.04

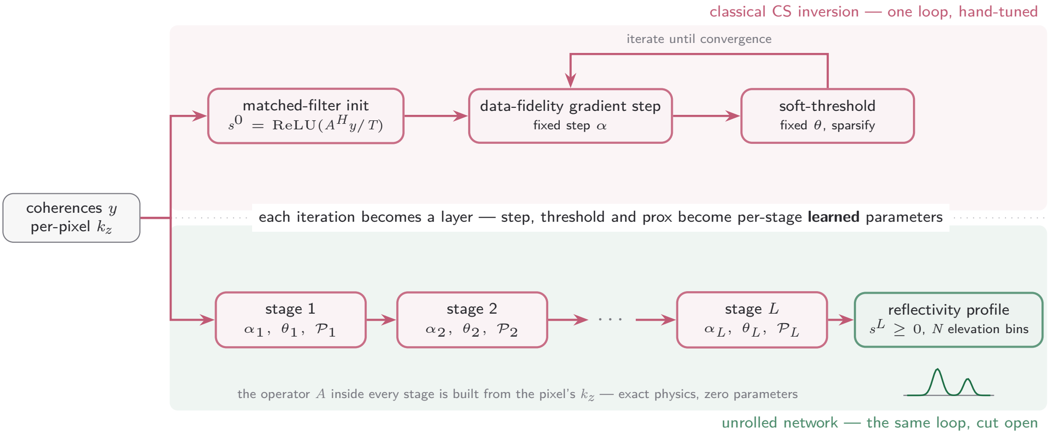
runs: dual UNet-skip trunks · set-pred · Hungarian · $K{=}2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Row = one physics term at weight w; param-L1 + cos = the no-physics reference. Five seeded replicas per arm, cells mean $\pm$ seed std; held-out test, 5.0 M px. **bold** = best in column, marked only where best and worst differ by more than 5% — no column clears it: curve fidelity and structure are flat at every gentle weight, and only degrade on the absolute-scale collapse arms (covariance / total power at $w{\geq}0.5$: RMSE 0.217 $\rightarrow$ 0.227, SSIM range 0.951 $\rightarrow$ 0.944). Peak location is pinned everywhere: median 0.67 and p95 4.7 in every arm; the mean moves in the second decimal. peak in elevation units.

The $K{=}2$ physics per-pixel tail — descriptor and covariance terms cap the blow-ups

objective	per-pixel R^2 ↑ med	mean	p_5	p_1	min	per-pixel cosine ↑ med	mean	p_1
param-L1 + cos	0.888 ±.007	-1.2 ±.8	$+0.3$ ±.1	-14.9 ±11.4	-13.2k ±3.3k	0.965 ±.004	0.934 ±.003	0.588 ±.011
+ *coherence re-synthesis*								
$w{=}0.01$	0.889 ±.005	-1.0 ±.8	$+0.3$ ±.1	-13.5 ±9.2	-12.7k ±7.4k	0.966 ±.002	0.935 ±.002	0.590 ±.006
$w{=}0.05$	0.888 ±.004	-1.2 ±.9	$+0.3$ ±.1	-14.8 ±8.1	-11.9k ±8.0k	0.966 ±.001	0.935 ±.002	0.587 ±.008
$w{=}0.25$	0.891 ±.006	-0.9 ±.5	$+0.3$ ±.0	-12.4 ±4.0	-12.2k ±4.8k	0.966 ±.002	0.936 ±.001	0.599 ±.009
$w{=}0.5$	0.888 ±.010	-9.6 ±12.0	$+0.2$ ±.2	-70.2 ±80.2	-66.3k ±77.4k	0.965 ±.003	0.935 ±.002	0.598 ±.006
$w{=}1$	0.870 ±.049	-3.7 ±4.3	-0.9 ±2.6	-50.2 ±71.0	-22.5k ±11.5k	0.966 ±.002	0.936 ±.001	**0.601** ±.007
+ *covariance matching*								
$w{=}0.01$	0.889 ±.008	$+0.7$ ±.0	$+0.4$ ±.0	-0.7 ±.2	-4.3k ±0.2k	0.965 ±.002	0.934 ±.001	0.575 ±.013
$w{=}0.05$	0.889 ±.009	$+0.8$ ±.0	$+0.4$ ±.0	-0.5 ±.1	-4.2k ±0.2k	0.966 ±.002	0.934 ±.001	0.574 ±.007
$w{=}0.25$	0.884 ±.006	$+0.8$ ±.0	$+0.4$ ±.0	-0.4 ±.0	-4.6k ±0.1k	0.966 ±.002	0.932 ±.001	0.544 ±.014
$w{=}0.5$	0.880 ±.007	$+0.8$ ±.0	$+0.4$ ±.0	-0.4 ±.0	-4.6k ±0.0k	0.965 ±.002	0.931 ±.001	0.529 ±.011
$w{=}1$	0.882 ±.004	$+0.8$ ±.0	$+0.4$ ±.0	$-\mathbf{0.3}$ ±.0	-4.6k ±0.1k	0.965 ±.001	0.929 ±.001	0.510 ±.025
+ *moments*								
$w{=}0.01$	0.888 ±.008	-0.1 ±.5	$+0.3$ ±.1	-7.1 ±6.6	-6.2k ±2.2k	0.966 ±.002	0.935 ±.001	0.588 ±.004
$w{=}0.05$	0.888 ±.009	$+0.5$ ±.2	$+0.4$ ±.0	-2.2 ±1.1	-4.4k ±1.2k	0.966 ±.002	0.935 ±.001	0.584 ±.006
$w{=}0.25$	0.888 ±.008	$+0.7$ ±.0	$+0.4$ ±.0	-0.8 ±.2	-4.2k ±0.1k	0.965 ±.002	0.934 ±.001	0.578 ±.004
$w{=}0.5$	0.887 ±.007	$+0.8$ ±.0	$+0.4$ ±.0	-0.5 ±.0	-4.3k ±0.3k	0.966 ±.002	0.934 ±.001	0.570 ±.009
$w{=}1$	0.886 ±.006	$+\mathbf{0.8}$ ±.0	$+\mathbf{0.4}$ ±.0	-0.4 ±.0	-4.4k ±0.3k	0.966 ±.001	0.933 ±.001	0.562 ±.008
+ *total power*								
$w{=}0.01$	0.889 ±.009	$+0.5$ ±.2	$+0.3$ ±.0	-2.6 ±1.2	$-\mathbf{3.6}$k ±0.2k	0.965 ±.002	0.935 ±.002	0.584 ±.005
$w{=}0.05$	0.889 ±.007	$+0.7$ ±.0	$+0.4$ ±.0	-0.9 ±.4	-4.1k ±0.1k	0.966 ±.001	0.934 ±.001	0.579 ±.008
$w{=}0.25$	0.888 ±.007	$+0.8$ ±.0	$+0.4$ ±.0	-0.5 ±.3	-4.3k ±0.3k	0.965 ±.001	0.933 ±.001	0.563 ±.016
$w{=}0.5$	0.887 ±.009	$+0.8$ ±.0	$+0.4$ ±.0	-0.5 ±.3	-4.2k ±0.3k	0.965 ±.002	0.931 ±.002	0.551 ±.010
$w{=}1$	0.885 ±.011	$+0.7$ ±.0	$+0.4$ ±.0	-0.8 ±.5	-4.2k ±0.2k	0.963 ±.005	0.929 ±.005	0.529 ±.026

runs: dual UNet-skip trunks · set-pred · Hungarian · $K{=}2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack → both trunks) · parity twins. Row = one physics term at weight w; param-L1 + cos = the no-physics reference. Five seeded replicas per arm, cells mean ± seed std; held-out test, $5.0\,$M px. pixel R^2 min and its spread in thousands (k). **bold** = best in column, marked only where best and worst differ by more than 5%. The typical pixel never moves (median and cosine flat); the tail is the story: catastrophic pixels wreck the no-physics mean (-1.17, p_1 -14.9, min -13.2k); covariance caps the blow-ups at every weight, improving as w grows (mean $+0.8$ at $w{=}1$); moments and total power insure at low weight (best min -3.6k). Coherence is NOT tail insurance: worse than the baseline at $w{>}0.5$ (mean -9.6, min -66k) — it shapes coherences, not pixel-domain outliers.

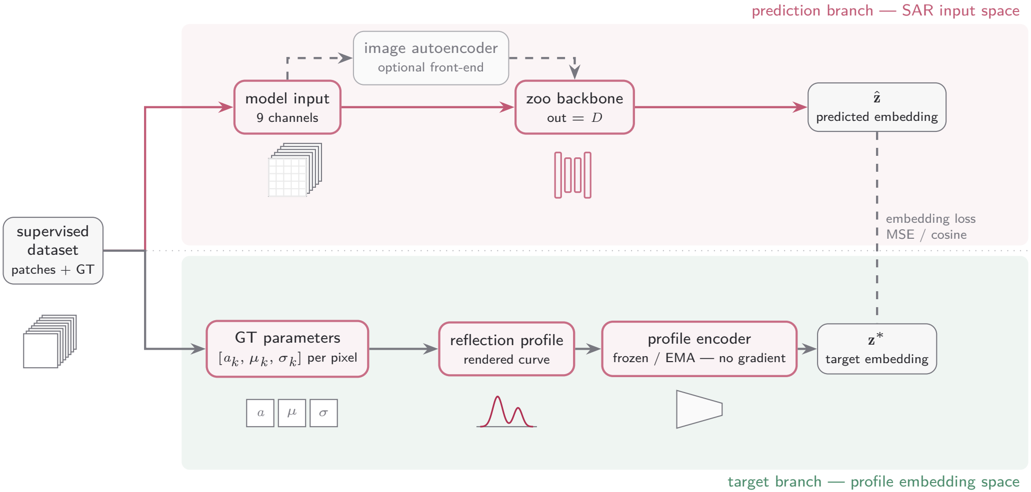
The unrolled model: the solver becomes the network



$$r^\ell = s^\ell + \alpha_\ell \,\mathrm{Re}\big[A^H(y - As^\ell)\big] / L_{\mathrm{lip}}, \qquad s^{\ell+1} = \mathrm{ReLU}\big(\mathcal{P}_\ell(r^\ell) - \theta_\ell\big)$$

γ-Net lineage (Qian et al., IEEE TGRS 2022); LISTA (Gregor & LeCun 2010); ISTA-Net (Zhang & Ghanem 2018). $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation; $L_{\mathrm{lip}} = TN\,\mathrm{d}z^2$ normalises the step. Default L=8: ~1.4k parameters.

What JEPA does



A network learns to predict, from the reduced stack alone, the **embedding** that a separately-trained profile autoencoder assigns to the ground-truth reflection profile.